

ASSIGNMENT

Unit – 1 INTRODUCTION TO PHP

Subject: Advanced Web Technology

Course: BCA – Semester 2 (GTU)

✓Basic HTML Questions

Q1. Explain basic HTML syntax with example. Why is HTML important in web development?

Answer:

HTML stands for **HyperText Markup Language**. It is the standard language used to create and design web pages. HTML provides the basic structure of a webpage by using different tags. These tags tell the browser how to display the content on the screen.

HTML syntax consists of opening and closing tags written inside angle brackets. Most HTML elements have a starting tag and an ending tag.

Example:

```
<!DOCTYPE html>
<html>
<head>
  <title>My Web Page</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is my first webpage.</p>
</body>
</html>
```

In this example, the <html> tag represents the root of the document. The <head> tag contains the title of the webpage. The <body> tag contains the actual content that is displayed in the browser such as headings and paragraphs.

HTML is very important in web development because it forms the foundation of every website. It helps in structuring web content and works together with CSS for styling and JavaScript for adding interactivity. Without HTML, it is not possible to create or display web pages in a browser.

Q2. Write short notes on HTML forms and explain how data is sent from browser to server.

Answer:

HTML forms are used to collect information from users such as name, email address, password and feedback. Forms allow users to interact with a website and submit their data to the server for processing.

An HTML form is created using the <form> tag. Inside the form, different input elements like text boxes, radio buttons and submit buttons are used.

Example:

```
<form action="process.php" method="post">  
  Name: <input type="text" name="username">  
  <input type="submit">  
</form>
```

When a user enters data in the form and clicks the submit button, the browser sends the entered data to the server. The data is sent to the file mentioned in the action attribute. The method attribute decides how the data is transferred, either using GET or POST method. On the server side, the data is processed using a server-side language such as PHP.

✔ PHP Syntax & Introduction

Q3. Explain basic PHP syntax with example and explain the use of PHP in web pages.

Answer:

PHP stands for **Hypertext Preprocessor**. It is a server-side scripting language used to create dynamic and interactive web pages. PHP code is executed on the server and the result is sent to the browser as plain HTML.

PHP code is written inside <?php and ?> tags. Each PHP statement ends with a semicolon. Variables in PHP start with a dollar sign.

Example:

```
<?php  
echo "Hello PHP";  
?>
```

PHP is used in web pages to perform various tasks such as processing form data, connecting to databases, generating dynamic content, managing sessions and cookies, and validating user input. Due to its simplicity and flexibility, PHP is widely used in web development.

Q4. What are variables in PHP? Explain different types of variables with examples.

Answer:

Variables in PHP are used to store data values that can be used and modified during program execution. A variable name always starts with a dollar sign followed by the variable name.

Different types of variables can store different types of data.

Examples:

```
$age = 20;           // Integer  
$price = 99.50;      // Float  
$name = "Payal";     // String  
$isStudent = true;   // Boolean  
$colors = array("Red", "Blue", "Green"); // Array  
$x = NULL;           // NULL
```

Each variable type is used according to the nature of data being stored. For example, integers are used for numbers, strings are used for text, and arrays are used to store multiple values in a single variable.

✓ Form Handling

Q5. Explain GET and POST method in detail with example.

Answer:

The GET and POST methods are used to send form data from the browser to the server.

The GET method sends data through the URL. The form data becomes visible in the browser's address bar. Due to this reason, GET is not secure and is generally used for non-sensitive data.

GET Example:

```
<form method="get" action="test.php">
```

```
<input type="text" name="name">
<input type="submit">
</form>
```

```
<?php
echo $_GET['name'];
?>
```

The POST method sends data in the request body, which is not visible in the URL. It is more secure than GET and is used when sensitive data like passwords are involved.

POST Example:

```
<form method="post" action="test.php">
  <input type="text" name="name">
  <input type="submit">
</form>
```

```
<?php
echo $_POST['name'];
?>
```

Q6. Explain how form data is received and processed in PHP with example.

Answer:

Form data in PHP is received using special variables called superglobal arrays such as `$_GET` and `$_POST`. These arrays store the data submitted through forms.

First, an HTML form is created to collect user input. When the user submits the form, the browser sends the data to the server. PHP then accesses the data using the name attributes of the form fields.

Example:

```
<form method="post" action="process.php">
  Email: <input type="text" name="email">
  <input type="submit">
</form>
```

```
<?php
$email = $_POST['email'];
echo "Entered Email: " . $email;
?>
```

✓ Number & String Handling

Q7. Explain arithmetic operations and number functions in PHP.

Answer:

PHP supports various arithmetic operations such as addition, subtraction, multiplication and division. These operations are used to perform calculations on numeric values.

Example:

```
$a = 20;  
$b = 10;  
echo $a + $b;  
echo $a - $b;  
echo $a * $b;  
echo $a / $b;
```

PHP also provides built-in number functions. The `rand()` function is used to generate a random number, while the `number_format()` function is used to format numbers with decimal points and commas.

```
echo rand(1, 50);  
echo number_format(12345.678, 2);
```

Q8. Explain string handling functions in PHP with examples.

Answer:

Strings in PHP are used to store and manipulate text. PHP provides many string functions to work with text data.

String concatenation is used to join two or more strings together. The dot (.) operator is used for this purpose.

```
echo "Hello " . "World";
```

The `strpos()` function is used to find the position of a substring within a string. The `str_replace()` function is used to replace a part of a string with another string.

```
echo strpos("Hello World", "World");  
echo str_replace("World", "PHP", "Hello World");
```

Control Structures

Q9. Explain if-else, switch and for loop in PHP with examples.

Answer:

Control structures in PHP are used to control the flow of program execution.

The if-else statement is used to execute different blocks of code based on a condition.

```
$age = 18;
if ($age >= 18) {
    echo "Eligible to vote";
} else {
    echo "Not eligible";
}
```

The switch statement is used when multiple conditions are based on a single variable.

```
$day = 1;
switch ($day) {
    case 1: echo "Monday"; break;
    case 2: echo "Tuesday"; break;
    default: echo "Invalid Day";
}
```

The for loop is used to repeat a block of code a specific number of times.

```
for ($i = 1; $i <= 5; $i++) {
    echo $i;
}
```

📌 IMPORTANT SHORT ANSWER QUESTIONS

1) Use of echo and print in PHP

The echo and print statements are used to display output on the screen. The echo statement can display multiple values and is faster than print. The print statement returns a value of 1, which can be used in expressions.

2) Difference between GET and POST methods

The GET method sends data through the URL, making it visible and less secure. The POST method sends data securely in the request body and is preferred for sensitive information.

3) PHP Comments and their types

Comments in PHP are used to explain the code and make it more readable. PHP supports single-line comments using `//` or `#` and multi-line comments using `/* */`.

4) Explain `rand()` and `number_format()` functions

The `rand()` function is used to generate a random number within a specified range. The `number_format()` function is used to format numbers by adding commas and decimal places.